Software Requirements Specification (SRS)

StayEase

Hostel & PG Booking Management System

1. Introduction

1.1 Purpose and Scope

This SRS describes StayEase, a web-based Hostel & PG Booking Management System. It supports public room browsing, customer registration and booking, and administrator management of rooms and reservations.

1.2 Intended Users

- Visitors: browse room listings and details.
- Customers: register, log in, check availability, manage bookings, and cancel eligible reservations.
- Administrators: manage room inventory, review reservations, approve or reject requests, and export booking data.

1.3 Terms

SRS: Software Requirements Specification; PG: Paying Guest; RBAC: Role-Based Access Control; CSV: Comma-Separated Values.

2. Overall Description

2.1 Technology and Environment

Component	Technology / Purpose
Language and framework	Java 21, Spring Boot 3.3.4
Web layer	Spring MVC and Thymeleaf templates
Security	Spring Security, role-based access, password hashing
Persistence	Spring Data JPA, Hibernate, MySQL 8.x
Frontend	HTML, Bootstrap, custom CSS
Testing	JUnit 5, MockMvc, AssertJ, H2 test dependency

2.2 Main Functions

- Browse, filter, and paginate room listings; view room details.
- Register and authenticate users, with separate customer and administrator access.
- Search availability and submit room booking requests.
- View booking history and cancel eligible bookings.
- Manage rooms and approve or reject reservations through the administrator portal.
- Export booking information as a CSV report.

3. Functional Requirements

ID	Requirement
FR-01	The system shall show public room listings, filters, pagination, and room details.
FR-02	The system shall allow visitors to register and users to log in using email and password.

FR-03	The system shall enforce role-based access to customer and administrator functions.
FR-04	The system shall validate room, guest count, and check-in/check-out dates before creating a booking.
FR-05	The system shall detect date conflicts with other PENDING or APPROVED bookings for the same room.
FR-06	The system shall save valid new booking requests with PENDING status.
FR-07	Customers shall view only their own bookings and cancel only eligible bookings they own.
FR-08	Administrators shall add, view, edit, activate/deactivate, and safely manage room records.
FR-09	Administrators shall approve or reject pending bookings; rejection shall require a remark.
FR-10	The system shall recheck availability during approval and coordinate concurrent booking attempts.
FR-11	Authorized administrators shall be able to export booking data to CSV.
FR-12	The system shall validate submitted data and display clear success, error, and confirmation messages.

4. External Interface Requirements

- User interface: responsive browser pages, forms, room listings, dashboards, booking tables, validation messages, and confirmation prompts.
- Software interfaces: Java/Spring Boot, Thymeleaf, Spring Security, Spring Data JPA/Hibernate, and MySQL.
- Communication: browser-to-server web requests and application-to-database JDBC connections.
- Hardware: no specialized hardware is required; users need a modern web browser.

5. Non-Functional Requirements

ID	Requirement
NFR-01 Security	Passwords shall be stored as hashes; protected routes shall enforce authorization and session controls.
NFR-02 Integrity	Booking dates, room capacity, ownership, and booking status shall be validated on the server.
NFR-03 Concurrency	Booking creation and approval shall safely coordinate competing requests for the same room.
NFR-04 Usability	Pages and error messages shall be clear and usable on desktop and mobile browsers.
NFR-05 Maintainability	Code shall remain separated into controllers, services, repositories, models, configuration, and templates.
NFR-06 Performance	Room and booking lists shall use pagination where supported.
NFR-07 Export safety	CSV output shall mitigate spreadsheet formula injection.
NFR-08 Configuration	Database credentials and administrator secrets shall be configured outside source code in deployment environments.

6. Data Requirements

Entity	Main attributes	Purpose
User	id, fullName, email, passwordHash, phone, role, createdAt	Account and access role.
Room	id, roomNumber, type, capacity, pricePerNight, description, active	Room inventory and pricing.
Booking	id, user, room, checkIn, checkOut, guests, status, adminRemark, createdAt	Reservation and its current status.

Relationships: one user may have many bookings; one room may be referenced by many bookings over time; each booking belongs to one user and one room.

7. Business Rules

7.1 Booking Overlap

Two bookings for the same room overlap when:

(Existing.checkIn < New.checkOut) AND (Existing.checkOut > New.checkIn)

Check-out is excluded from the stay interval, so one guest may check out on the same date another checks in.

7.2 Status and Validation Rules

- PENDING and APPROVED bookings block dates; REJECTED and CANCELLED bookings do not.
- Check-out must be later than check-in, the room must be active, and guest count must not exceed room capacity.
- New booking requests start as PENDING.
- Administrators must recheck availability before approval and provide a remark when rejecting a booking.
- Customers may cancel only their own eligible PENDING or APPROVED bookings, subject to the application's future-stay rule.
- Transactions and a room-row pessimistic lock are used in booking creation/approval to reduce double-booking race conditions.

8. Main Use Cases

Use case	Actor	Outcome
Browse rooms	Visitor	Views and filters room listings and details.
Register / log in	Visitor / user	Creates an account or enters the role-appropriate portal.
Create booking	Customer	A valid, conflict-free request is saved as PENDING.
Manage booking	Customer	Views own booking history and cancels an eligible booking.
Manage rooms	Administrator	Creates or updates rooms and changes active status.
Review booking	Administrator	Approves or rejects a request; rejection includes a remark.
Export report	Administrator	Downloads supported booking data in CSV format.

9. Assumptions and Constraints

- The environment has Java 21, a compatible MySQL server, and required Maven dependencies.
- The application must be configured with valid database connection details and an authorized administrator account.
- Payment gateway integration is outside the confirmed scope of this SRS.
- Email notifications, seasonal pricing, and room-maintenance features are mentioned in descriptive material but were not confirmed as implemented in the reviewed source.
- The README reports 178 passing tests; independent execution was blocked in the review environment by unavailable Maven repository network/DNS access.

10. Acceptance Criteria

- Visitors can browse room listings and details without signing in.
- Registration and login validate user input and apply role-based access.
- The system rejects invalid dates, over-capacity requests, inactive rooms, and overlapping blocking bookings.
- Same-day checkout/check-in boundaries are allowed when no other conflict exists.
- Customers can access and cancel only their own eligible bookings.
- Administrators can manage rooms and review bookings; rejection requires a remark.
- Concurrent booking attempts cannot both successfully create conflicting reservations through the protected workflow.
- Authorized administrators can export CSV data with formula-injection mitigation.

11. References

1. ISO/IEC/IEEE 29148:2018 — Systems and Software Engineering: Requirements Engineering.
2. Sommerville, Ian. Software Engineering, 10th Edition, Pearson.
3. Oracle — Java Documentation.
4. Spring — Spring Boot Reference Documentation.
5. Spring — Spring Security Reference Documentation.
6. Spring — Spring Data JPA Reference Documentation.
7. Thymeleaf — Official Documentation.